ECDAT Quantum Cryptographic Risk Report

Target: **ECDAT Vulnerable Benchmark App** | Generated: 2026-11-08 19:58 UTC

Total Assets	At Risk (Mosca)	Quantum Vulnerable	Critical / High
45	23	37.8%	24

Quantum Risk Horizon (Mosca's Theorem: X + Y > Z)

Under the *Harvest Now, Decrypt Later* threat, encrypted records with shelf-life **X** migrating over **Y** years are at risk if **X + Y > Z** (estimated CRQC horizon ~2038).

Top Urgent Cryptographic Risks

Asset	Score	Criticality	File	Threat Analysis
TLS 1.0	**80.0**	LOW	.env	TLS 1.0 is classically broken and obsolete; vulnerable to immediate exploitation today regardless of quantum timeline.
RC4	**80.0**	LOW	.env	RC4 is classically broken and obsolete; vulnerable to immediate exploitation today regardless of quantum timeline.
TLS 1.0	**80.0**	LOW	nginx.conf	TLS 1.0 is classically broken and obsolete; vulnerable to immediate exploitation today regardless of quantum timeline.
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3DES	**80.0**	LOW	nginx.conf	3DES is classically broken and obsolete; vulnerable to immediate exploitation today regardless of quantum timeline.
MD5	**80.0**	LOW	nginx.conf	MD5 is classically broken and obsolete; vulnerable to immediate exploitation today regardless of quantum timeline.

Prioritized Post-Quantum Migration Action Plan

Primitive	PQC Replacement	Approach	Effort	Implementation
TLS 1.0	**TLS 1.3 with Hybrid Post-Quantum Key Exchange (X25519MLKEM768)**	HYBRID	MEDIUM	OpenSSL 3.5+, BoringSSL
RC4	**AES-256-GCM (NIST SP 800-38D) or ChaCha20-Poly1305 (RFC 8439)**	CLASSICAL_UPGRADE	LOW	Standard library, cryptography
TLS 1.0	**TLS 1.3 with Hybrid Post-Quantum Key Exchange (X25519MLKEM768)**	HYBRID	MEDIUM	OpenSSL 3.5+, BoringSSL
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3DES	**AES-256-GCM (NIST SP 800-38D) or ChaCha20-Poly1305 (RFC 8439)**	CLASSICAL_UPGRADE	LOW	Standard library, cryptography
MD5	**SHA-256 or SHA-3 (NIST FIPS 202) / SHA-384**	CLASSICAL_UPGRADE	LOW	Standard library hashlib, crypto